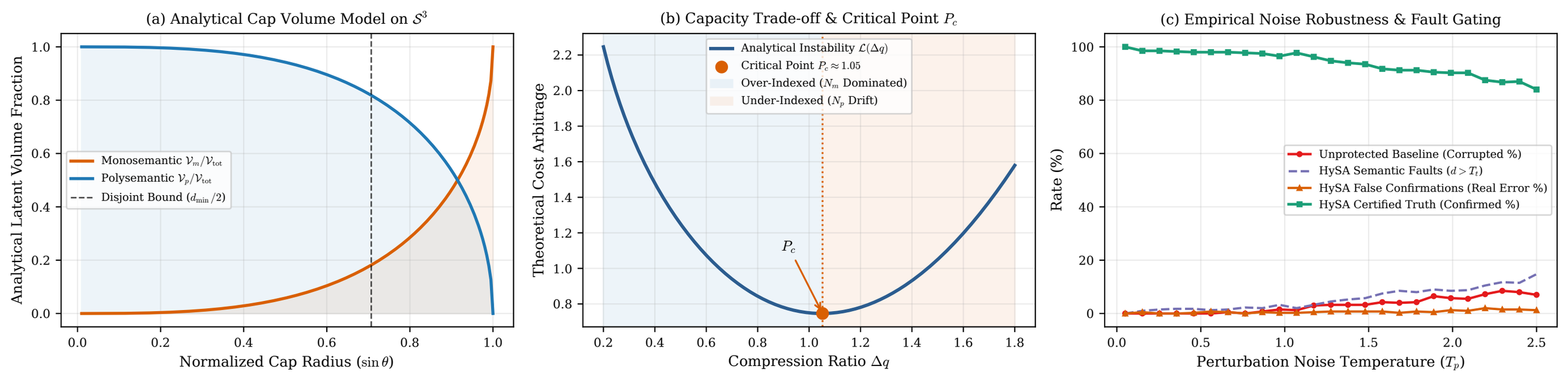
(a) Analytical Cap Volume Model on $\mathcal{S}^3$
Analytical Latent Volume Fraction
Normalized Cap Radius ($\sin\theta$)
Monosemantic $\mathcal{V}_m/\mathcal{V}_{\text{tot}}$
Polysemantic $\mathcal{V}_p/\mathcal{V}_{\text{tot}}$
Disjoint Bound ($d_{\min}/2$)
(b) Capacity Trade-off & Critical Point P_c
Theoretical Cost Arbitrage
Compression Ratio Δq
Analytical Instability $\mathcal{L}(\Delta q)$
Critical Point $P_c \approx 1.05$
Over-Indexed (N_m Dominated)
Under-Indexed (N_p Drift)
P_c
(c) Empirical Noise Robustness & Fault Gating
Rate (%)
Perturbation Noise Temperature (T_p)
Unprotected Baseline (Corrupted %)
HySA Semantic Faults ($d > T_t$)
HySA False Confirmations (Real Error %)
HySA Certified Truth (Confirmed %)